# Homework Assignment 1: Slab-Geometry Boltzmann Equation

Tal Ramot

August 15, 2026

## 1 Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x, \mu, t; c) = ce^{-(1-c)v\Sigma_t t}\psi(cx, \mu, ct; 1) \tag{1}$$

We will do it in two parts. First, we will write down the equation that $\tilde{\psi} = \psi(cx, \mu, ct; 1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)v\Sigma_t t}\tilde{\psi}$ satisfies the same equation as $\psi(x, \mu, t; c)$.

### 1.1 The equation for $\tilde{\psi}$

**Notation:** We will denote by $\Psi(X, \mu, T) \equiv \psi(X, \mu, T; 1)$ the $c = 1$ solution written in its own variables, and by $\Psi_T$ and $\Psi_X$ its partial derivatives with respect to them, so that $\tilde{\psi}(x, \mu, t) = \Psi(cx, \mu, ct)$ with $X = cx$ and $T = ct$. **This notation is used to avoid confusion when using the chain rule, as will be done immediately.** First, $\Psi$ satisfies the $c = 1$ equation in the variables $X$ and $T$:

$$\frac{1}{v}\Psi_T + \mu\Psi_X + \Sigma_t\Psi = \frac{\Sigma_t}{2}\int_{-1}^{1}\Psi d\mu' \tag{2}$$

which after applying the chain rule, $\Psi_T = \frac{1}{c}\frac{\partial\tilde{\psi}}{\partial t}$ and $\Psi_X = \frac{1}{c}\frac{\partial\tilde{\psi}}{\partial x}$, and multiplying by $c$, becomes:

$$\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + c\Sigma_t\tilde{\psi} = \frac{c\Sigma_t}{2}\int_{-1}^{1}\tilde{\psi}d\mu' \tag{3}$$

So $\tilde{\psi}$ obeys the same equation as $\psi(x, \mu, t; c)$ except that it is removed at the rate $c\Sigma_t$ instead of $\Sigma_t$.

### 1.2 Proving the identity

In this part we are going to simply subtitute the RHS of 1 into the LHS of the equation for $\psi(x, \mu, t; c)$ and show that it satisfies the same equation. After substitution and application of the chain rule we get:

$$\text{LHS} = \frac{1}{v}\left[c(c-1)v\Sigma_t e^{-(1-c)v\Sigma_t t}\tilde{\psi} + c\,e^{-(1-c)v\Sigma_t t}\frac{\partial\tilde{\psi}}{\partial t}\right] + \mu c\,e^{-(1-c)v\Sigma_t t}\frac{\partial\tilde{\psi}}{\partial x} + \Sigma_t c\,e^{-(1-c)v\Sigma_t t}\tilde{\psi} \quad (4)$$

$$= ce^{-(1-c)v\Sigma_t t}\left[\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + \Sigma_t\tilde{\psi} + (c-1)\Sigma_t\tilde{\psi}\right] \quad (5)$$

$$= ce^{-(1-c)v\Sigma_t t}\left[\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + c\Sigma_t\tilde{\psi}\right] \quad (6)$$

$$= ce^{-(1-c)v\Sigma_t t}\frac{c\Sigma_t}{2}\int_{-1}^{1}\tilde{\psi}d\mu' = \frac{c\Sigma_t}{2}\int_{-1}^{1}ce^{-(1-c)v\Sigma_t t}\tilde{\psi}d\mu' \quad (7)$$

where 3 was used in the last line, and this is exactly the RHS of the equation for $\psi(x,\mu,t;c)$.

## 2 Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi\int_{|x|}^{\infty}\phi_{\text{pt}}(r)rdr \quad (8)$$

The solution taken from J. C. J. Paasschens is the point-source scalar flux $\phi_{\text{pt}}(r,t)$ for $c=1$, and therefore the planar-source scalar flux is given by the integration given above.

$$\phi_{\text{pl}}(x,t) = 2\pi\int_{|x|}^{\infty}\frac{e^{-v\Sigma_t t}}{4\pi r^2}\delta(r-vt)rdr + 2\pi\int_{|x|}^{\infty}\frac{\left(1-r^2/v^2t^2\right)^{1/8}}{\left(4\pi vt/(3\Sigma_t)\right)^{3/2}}e^{-v\Sigma_t t}G\left(v\Sigma_t t\left[1-\frac{r^2}{v^2t^2}\right]^{3/4}\right)\Theta(vt-r)rdr \quad (9)$$

$$= \frac{e^{-v\Sigma_t t}}{2vt}\Theta(vt-|x|) + \Theta(vt-|x|)\frac{2\pi e^{-v\Sigma_t t}}{\left(4\pi vt/(3\Sigma_t)\right)^{3/2}}\int_{|x|}^{vt}\left(1-\frac{r^2}{v^2t^2}\right)^{1/8}G\left(v\Sigma_t t\left[1-\frac{r^2}{v^2t^2}\right]^{3/4}\right)rdr \quad (10)$$

$$\overset{\xi=r/vt}{=} \frac{e^{-v\Sigma_t t}}{2vt}\Theta(vt-|x|) + \Theta(vt-|x|)\frac{2\pi e^{-v\Sigma_t t}(vt)^{1/2}}{\left(4\pi/(3\Sigma_t)\right)^{3/2}}\int_{\frac{|x|}{vt}}^{1}\left(1-\xi^2\right)^{1/8}G\left(a\left[1-\xi^2\right]^{3/4}\right)\xi d\xi \quad (11)$$

$$\overset{\substack{u=1-\xi^2\\ w=au^{3/4}}}{=} \frac{e^{-v\Sigma_t t}}{2vt}\left[1+\sqrt{\frac{3}{\pi}}\int_0^{w_0}\sqrt{w}G(w)dw\right]\Theta(vt-|x|) \quad (12)$$

with $w_0 = a\left(1-\frac{x^2}{(vt)^2}\right)^{3/4}$, the powers combining as $\frac{1}{6}+\frac{1}{3}=\frac{1}{2}$ and the $\Sigma_t$ dependence cancelling in the last step (I used the assistance of AI in this algebric derivation). Finally, with the interpolation formula of Paasschens, $G(w) = e^w\sqrt{1+b/w}$ with $b=2.026$, the integral has a closed form:

$$\int_0^{w_0}e^w\sqrt{w+b}\,dw = e^{w_0}\sqrt{w_0+b} - \sqrt{b} - \frac{\sqrt{\pi}}{2}e^{-b}\left[\text{erfi}\left(\sqrt{w_0+b}\right) - \text{erfi}\left(\sqrt{b}\right)\right] \quad (13)$$

which **closes the derivation of the planar-source scalar flux for $c=1$.** Of course the relation proved in Q1 can be used to get the solution for $c \neq 1$ using a simple scaling and multiplication by an exponential factor of the solution for $c=1$.

# 3 Question 3 - Classical and asymptotic diffusion against the exact solution

## 3.1 The three solutions being compared

Throughout, $\Sigma_t = v = 1$ as specified in the assignment, the source is the plane pulse $Q(x,t) = \delta(x)\delta(t)$, and the figures are generated by `src/homework2`.

This section covers parts 3(a) and 3(b): all three curves being compared are closed forms, so no numerical scheme enters. Parts 3(c) and 3(d) — the finite-difference time-dependent diffusion code and the relative-error study — are separate deliverables, and the analytic solution derived below is what that code will be verified against.

The source being a pulse in *time* as well as in space is not an assumption but a requirement of the solution being used: Paasschens takes the isotropic point source $S(\mathbf{r}, t, \hat{\mathbf{s}}) = \delta(\mathbf{r})\delta(t)$ (his Eq. 4), and the uncollided term $\propto \delta(r - vt)$ is its signature — every uncollided particle sits at radius exactly $vt$, which only holds for an instantaneous release. A steady $\delta(\mathbf{r})$ source would give a time-independent $e^{-\Sigma_t r}/(4\pi r^2)$ instead, and there would be no $t = 1, 2, 3, 4, 7, 15$ to present.

The **exact** solution is the planar-source scalar flux derived in Question 2 from the Paasschens point-source solution, carried to general $c$ by the scaling identity of Question 1:

$$\phi(x,t;1) = \frac{e^{-t}}{2t}\left[1 + \sqrt{\frac{3}{\pi}}\int_0^{w_0} \sqrt{w}\, G(w)\, dw\right]\Theta(t - |x|), \qquad w_0 = t\left(1 - \frac{x^2}{t^2}\right)^{3/4} \tag{14}$$

$$\phi(x,t;c) = c\, e^{-(1-c)t}\, \phi(cx, ct; 1) \tag{15}$$

The two **diffusion** solutions are the Green's functions of

$$\frac{1}{v}\frac{\partial \phi}{\partial t} - D\frac{\partial^2 \phi}{\partial x^2} + \Sigma_a \phi = \delta(x)\delta(t), \qquad \Sigma_a = (1-c)\Sigma_t, \tag{16}$$

derived in Section 3.2, differing only through $D = D_0(c)/\Sigma_t$:

$$D_0 = \tfrac{1}{3} \quad \text{(classical)}, \qquad D_0(c) = (1-c)\nu_0^2 \quad \text{(asymptotic)}, \tag{17}$$

where $\nu_0$ is the discrete transport eigenvalue, $c\nu_0 \operatorname{arctanh}(1/\nu_0) = 1$. For $c > 1$ the eigenvalue is imaginary, $\nu_0 = i|\nu_0|$, and $(1-c)\nu_0^2 = (c-1)|\nu_0|^2$ remains positive; both branches tend to $\frac{1}{3}$ as $c \to 1$, so the two approximations coincide there.

| $c$ | 0.6 | 0.8 | 1.0 | 1.2 | 1.5 |
|---|---|---|---|---|---|
| $D_0(c)$ | 0.48588 | 0.39629 | 0.33333 | 0.28717 | 0.23745 |

Table 1: The asymptotic diffusion coefficient; the classical value is 1/3 for every $c$.

## 3.2 Derivation of the time-dependent diffusion Green's function

### 3.2.1 The equation, and where the $1/v$ belongs

Written for the *scalar flux* $\phi = vn$, the time-dependent diffusion equation carries a factor $1/v$ on the time derivative:

$$\frac{1}{v}\frac{\partial \phi}{\partial t} - D\frac{\partial^2 \phi}{\partial x^2} + \Sigma_a \phi = S(x,t) \tag{18}$$

The quickest way to see that it must be there is dimensional. With $D = D_0/\Sigma_t$ a length and $\Sigma_a$ an inverse length, both $D\,\partial_x^2\phi$ and $\Sigma_a\phi$ carry units of $\phi$/length; the time derivative carries $\phi$/time, and only the factor $1/v$, of units time/length, brings it into line. Equivalently, $\partial_x$ and $\frac{1}{v}\partial_t$ are the pair that appear together in the transport equation from which (18) descends, since a particle covers $v\,dt$ of path in $dt$ of time.

Written instead for the *density* $n = \phi/v$, the same equation reads $\partial_t n - Dv\,\partial_x^2 n + v\Sigma_a n = vS$, with no $1/v$: the factor is a statement about which variable is being used, not an extra physical ingredient. In the dimensionless units of this assignment, $\Sigma_t = v = 1$, the two forms coincide, which is why the distinction never reaches the numbers.

### 3.2.2 Reduction to the heat equation

Take $S = \delta(x)\delta(t)$ and multiply (18) by $v$:

$$\frac{\partial\phi}{\partial t} - Dv\frac{\partial^2\phi}{\partial x^2} + v\Sigma_a\phi = v\,\delta(x)\delta(t) \tag{19}$$

**Step 1: the pulse source is an initial condition.** Integrate (19) over $t \in (0^-, 0^+)$. The diffusion and absorption terms are bounded and contribute nothing in the limit, so only the time derivative survives:

$$\phi(x,0^+) - \phi(x,0^-) = v\,\delta(x) \qquad \Longrightarrow \qquad \phi(x,0^+) = v\,\delta(x) \tag{20}$$

since nothing precedes the pulse. For $t > 0$ the equation is source-free, so the problem is now a pure initial-value problem. This is the same manoeuvre as in Assignment 1, Question 2, where integrating the steady equation across the source turned the delta into a boundary current instead: a delta source is always better absorbed into a condition than kept as a term.

**Step 2: absorption factors out exactly.** Absorption removes particles at a rate independent of position, so it can only rescale the solution, never reshape it. Substituting $\phi(x,t) = e^{-v\Sigma_a t}\,u(x,t)$ into (19) for $t > 0$, the $-v\Sigma_a u$ produced by the time derivative cancels the absorption term identically and leaves the pure heat equation

$$\frac{\partial u}{\partial t} = Dv\frac{\partial^2 u}{\partial x^2}, \qquad u(x,0^+) = v\,\delta(x) \tag{21}$$

Note this step is untouched by the sign of $1-c$: for $c > 1$ the exponential grows instead of decays, and nothing else changes.

**Step 3: Fourier transform.** With $\hat{u}(k,t) = \int_{-\infty}^{\infty} u(x,t)e^{-ikx}dx$, the spatial derivative becomes multiplication by $-k^2$ and the PDE becomes an ordinary differential equation in $t$ for each $k$ separately:

$$\frac{d\hat{u}}{dt} = -Dvk^2\,\hat{u}, \qquad \hat{u}(k,0^+) = v \qquad \Longrightarrow \qquad \hat{u}(k,t) = v\,e^{-Dvk^2 t} \tag{22}$$

the initial value being $v$ because the transform of $v\delta(x)$ is the constant $v$ — a delta excites every wavenumber equally.

**Step 4: invert.** Completing the square in the exponent, $-Dvtk^2+ikx = -Dvt\left(k-\frac{ix}{2Dvt}\right)^2 - \frac{x^2}{4Dvt}$, and using $\int_{-\infty}^{\infty} e^{-\alpha k^2} dk = \sqrt{\pi/\alpha}$:

$$u(x,t) = \frac{v}{2\pi}\int_{-\infty}^{\infty} e^{-Dvk^2t} e^{ikx} dk = \frac{v}{\sqrt{4\pi Dvt}} e^{-x^2/(4Dvt)} \tag{23}$$

Restoring the factor from Step 2 gives the Green's function:

$$\boxed{\phi_{\text{diff}}(x,t;c) = \frac{v\, e^{-(1-c)\Sigma_t vt}}{\sqrt{4\pi Dvt}} \exp\left(-\frac{x^2}{4Dvt}\right)} \tag{24}$$

which in the units of this assignment, $\Sigma_t = v = 1$ and $D = D_0(c)$, is the form implemented in `src/homework2/diffusion.py`:

$$\phi_{\text{diff}}(x,t;c) = \frac{e^{-(1-c)t}}{\sqrt{4\pi D_0(c)\, t}} e^{-x^2/(4D_0(c)t)} \tag{25}$$

### 3.2.3 Three checks on the Green's function

**Particle balance.** Integrating (24) over all $x$ gives $\int \phi\, dx = v\, e^{-(1-c)\Sigma_t vt}$, i.e. $\int n\, dx = e^{-(1-c)t}$ in dimensionless units. This is exactly the normalisation the exact transport solution obeys, by the scaling identity (15) — so the two solutions being compared carry the same number of particles at every time, and the differences in the figures are differences of *shape* alone.

**Spreading rate.** The Gaussian has variance $\langle x^2\rangle = 2Dvt$, so the particle cloud spreads as $\sqrt{t}$ with no upper bound on speed. The exact solution spreads at most as $vt$. This is the origin of the leakage past the causal front seen in every panel of Section 3.3, and it is a structural defect of the diffusion approximation, not a small-parameter error.

**The steady limit.** A steady source is a pulse source fired repeatedly, so integrating (24) over all elapsed times must reproduce the steady solution. Using

$$\int_0^{\infty} \frac{e^{-\beta\tau}}{\sqrt{4\pi a\tau}} e^{-x^2/(4a\tau)} d\tau = \frac{1}{2\sqrt{a\beta}} e^{-\sqrt{\beta/a}\,|x|} \tag{26}$$

with $\tau = vt$, $a = D$ and $\beta = \Sigma_a$:

$$\int_0^{\infty} \phi_{\text{diff}}\, dt = \frac{1}{2\sqrt{D\Sigma_a}} e^{-\sqrt{\Sigma_a/D}\,|x|} \overset{D=1/(3\Sigma_t)}{=} \frac{\sqrt{3}}{2\sqrt{1-c}} e^{-\sqrt{3(1-c)}\,\Sigma_t|x|} \tag{27}$$

the familiar classical result, and with $D = (1-c)\nu_0^2/\Sigma_t$ instead, $e^{-\Sigma_t|x|/\nu_0}/\bigl(2(1-c)\nu_0\bigr)$, the asymptotic one. This ties the coefficient in front of the Gaussian to the coefficient in front of the exponential, so a normalisation slip in either shows up immediately; it is run numerically in `main.py` and agrees to $\sim 10^{-14}$ relative. Note it exists only for $c < 1$: for $c \geq 1$ nothing removes the particles a steady source keeps adding, the integral diverges, and only the time-dependent form (24) is meaningful.

## 3.3 Parts 3(a) and 3(b): the comparison

Each figure fixes $c$ and shows $t = 1, 2, 3, 4, 7, 15$. The vertical line marks the causal front $|x| = vt$, beyond which the exact solution is identically zero. Only $x \geq 0$ is plotted; all three solutions are even in $x$.

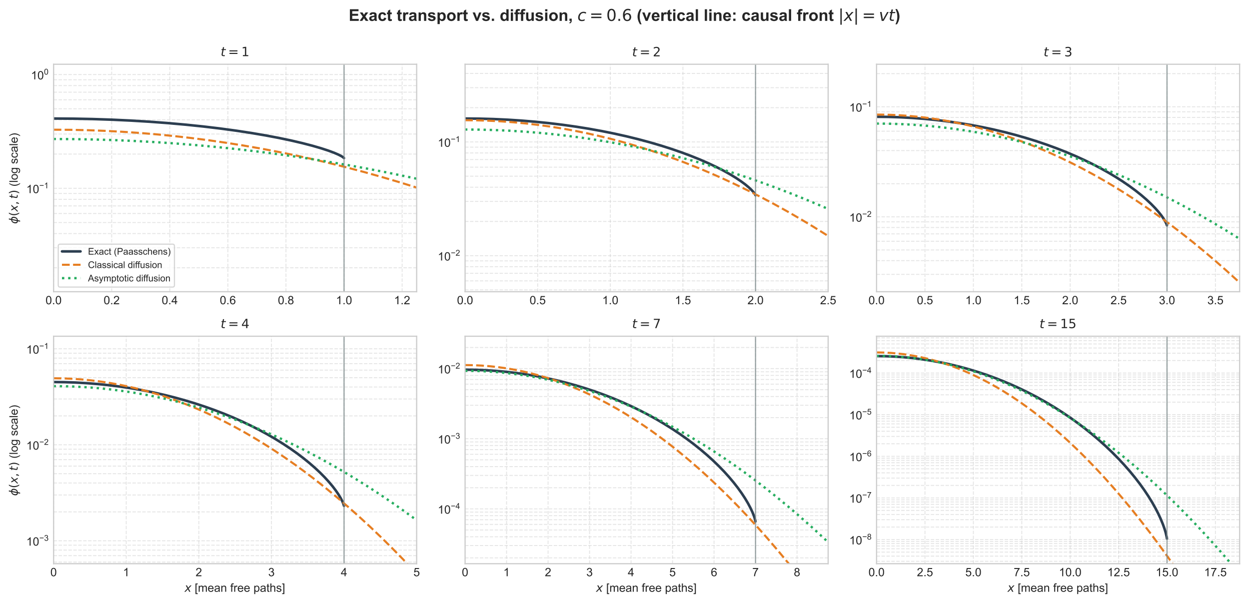



Figure 1: $c = 0.6$.

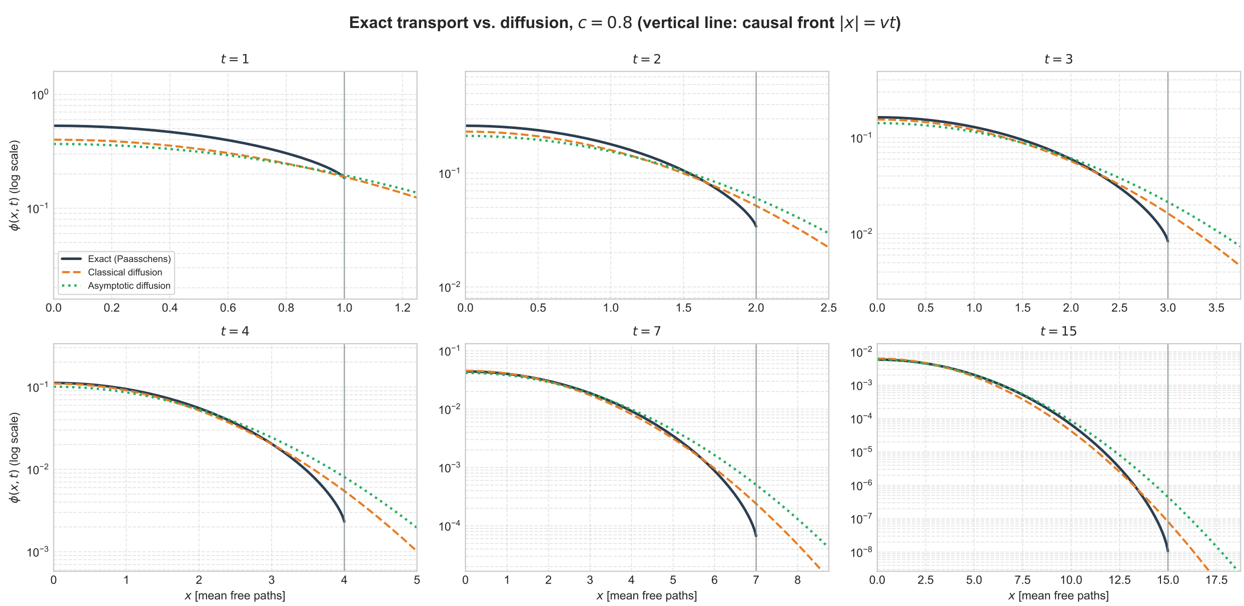



Figure 2: $c = 0.8$.

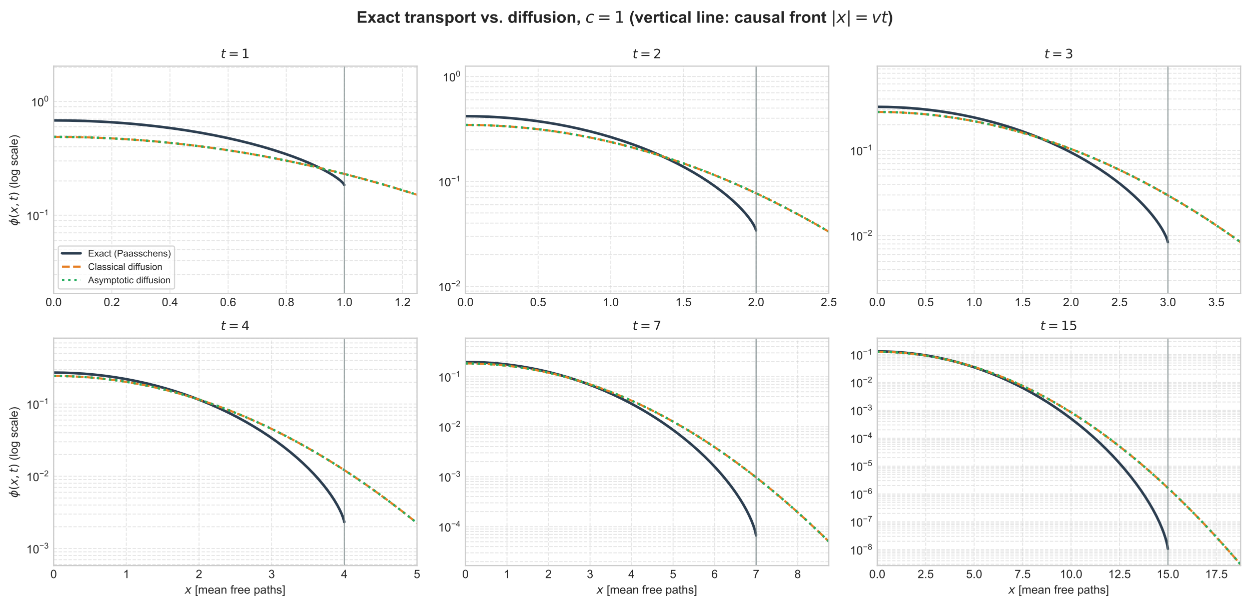



Figure 3: $c = 1$. The two diffusion coefficients coincide at $D_0 = 1/3$, so the classical and asymptotic curves overlie each other.

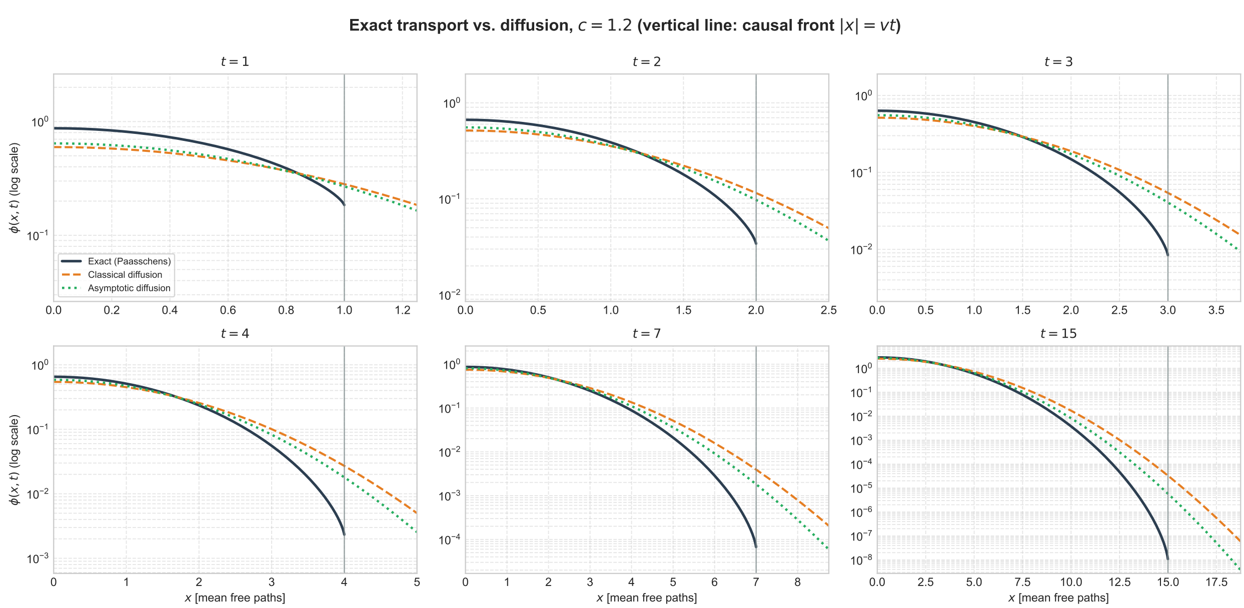
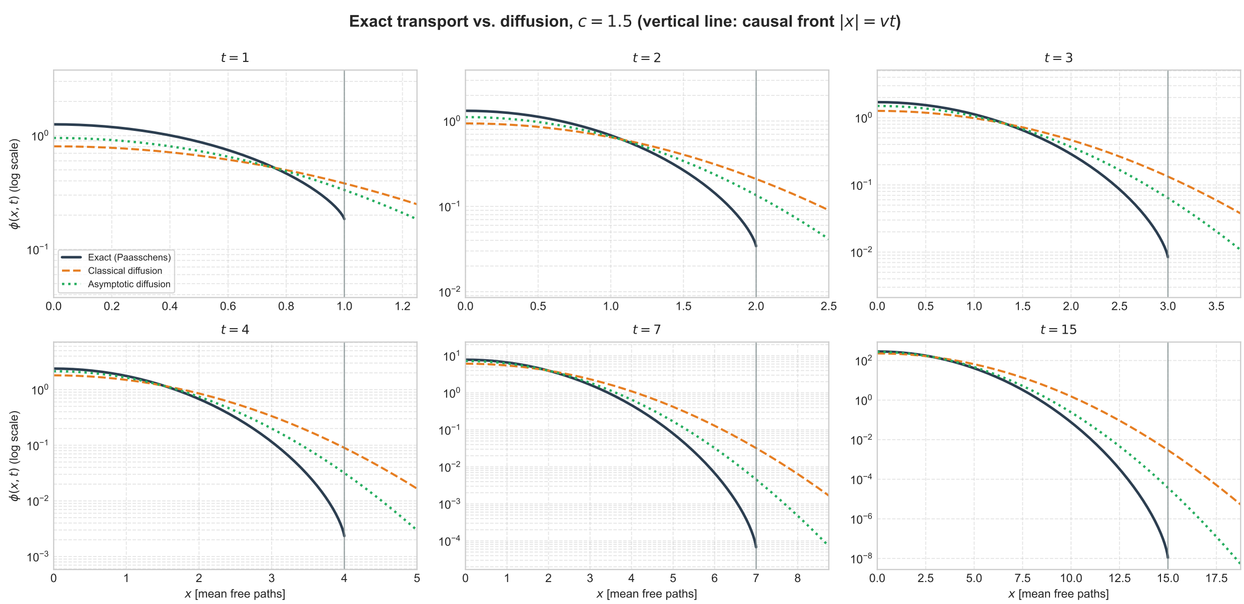



Figure 4: $c = 1.2$.



Exact transport vs. diffusion, $c = 1.5$ (vertical line: causal front $|x| = vt$)
$t = 1$
$t = 2$
$t = 3$
$t = 4$
$t = 7$
$t = 15$
$\phi(x, t)$ (log scale)
$x$ [mean free paths]
Exact (Paasschens)
Classical diffusion
Asymptotic diffusion


Figure 5: $c = 1.5$.